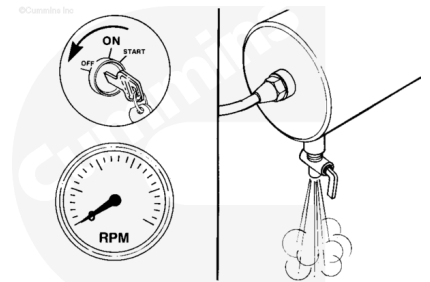


Trainings ▾

Initial Check

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

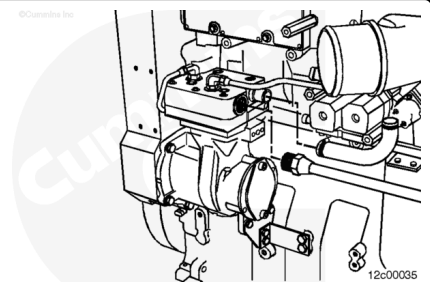


Shut the engine OFF.

Open the drain cock on the wet tank to release compressed air from the system.

⚠ WARNING ⚠

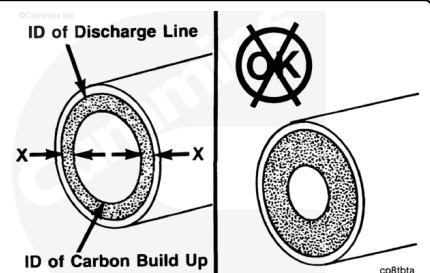
Air discharge lines can be very hot. Be sure the lines are cool before handling to prevent personal injury.



Remove the air inlet and outlet connections from the air compressor.

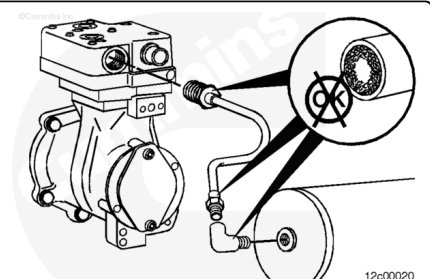
Measure the total carbon deposit thickness inside the air discharge line as shown.

Note : The carbon deposit thickness must not exceed 1.6 mm [0.06 in].



⚠ WARNING ⚠

The air discharge line must be capable of withstanding extreme heat and pressure to prevent personal injury and property damage. Refer to the OEM specifications.



Note : If the total carbon deposit thickness exceeds specification, remove and clean, or replace, the air

discharge line.

Continue to check for carbon buildup in the air discharge line connections up to the first connection or wet tank.

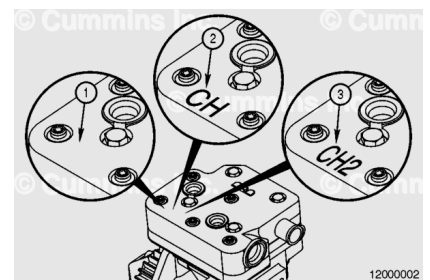
Clean or replace any lines and fittings with carbon deposits thicker than 1.6 mm [0.06 in]. Refer to the OEM specifications for cleaning or replacement instructions.

Preparatory Steps

- Check the marking on the air compressor head before cleaning or replacing the cylinder head.

Repair Direction:

1. No marking: Perform on-engine repair.
2. CH marking: Perform off-engine repair.
3. CH2 marking: Replace the air compressor cylinder head.
Refer to Procedure 012-007 in Section 12.
(</qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html>)



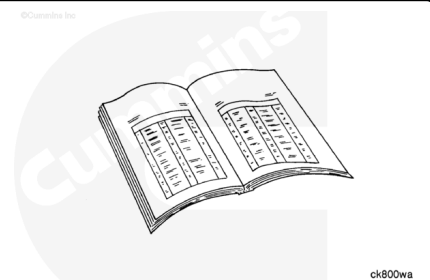
Clean

On-Engine Repair

Use the Air Compressor Cleaning Kit, Part Number 2892280, to clean and remove internal carbon deposits and other foreign material from the air compressor cylinder head.

The items below are **not** included in the air compressor cleaning kit, Part Number 2892280, but are required to perform the repair.

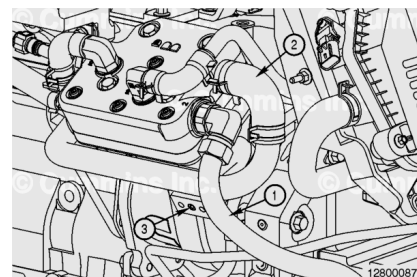
- Regulated shop air 621 kPa [90 psi]
- Automatic transmission fluid
- Shop rags
- 19 liter [5 gal] bucket
- INSITE™ electronic service tool.



Park vehicle on a level surface and in an area where it is safe to idle for an extended period of time. Chock the vehicle wheels, and completely drain the vehicle air system.



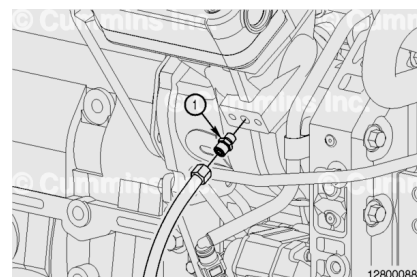
Remove the air compressor discharge line (1), air compressor intake line (2), and air governor signal line (3) at the air compressor.



Install and tighten the M10 connector (1) into the unloader port of the air compressor housing.

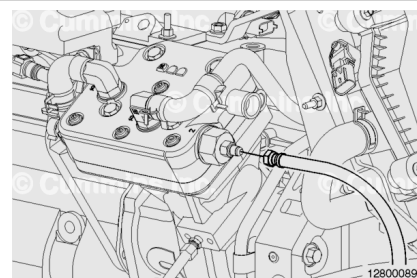
Install and tighten the unloader control valve hose to the M10 connector installed into the air compressor housing.

Note : Some air compressors have two unloader ports in the air compressor housing. Make sure that one of the ports is plugged with an appropriate pipe plug.



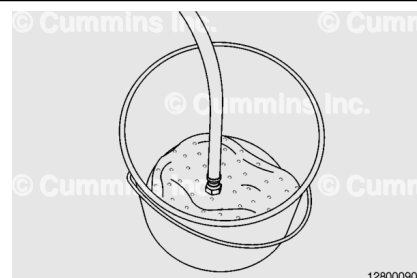
⚠ CAUTION ⚠

While cleaning the air compressor cylinder head, do NOT install the original intake or discharge plumbing. Only use the intake and discharge lines and fitting supplied with the air compressor cleaning kit during this procedure.



Install the discharge line and clamp from the kit, Part Number 2892280, onto the discharge port fitting on the air compressor cylinder head.

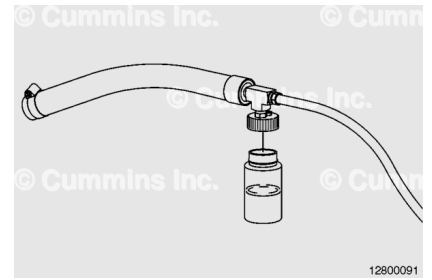
Place the discharge line into a 19 liter [5 gal] bucket and cover with several oil absorption pads.



Pour 89 ml [3 fl oz] of automatic transmission fluid into the dispensing bottle.

Install the dispensing bottle onto the dispensing hose assembly.

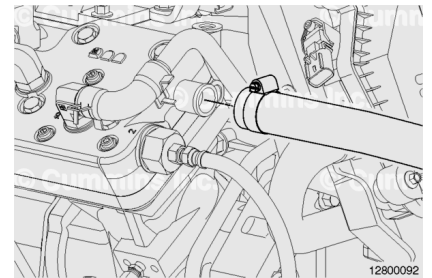
Note : For best results, use Dexron III automatic transmission fluid, or equivalent.



Install and tighten the dispensing hose assembly and clamp to the intake port of the air compressor cylinder head.

Connect the regulated shop air at 621 kPa [90 psi] to the air compressor cleaner.

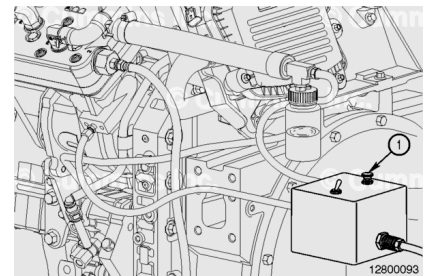
Note : The next three steps will happen quickly. For best results, read the remaining instructions prior to continuing with the cleaning procedure. Do not cycle the unloader until all 89 ml [3 fl oz] of automatic transmission fluid have been applied.



Start the engine and let it idle for 1 minute.

The air compressor may draw in the automatic transmission fluid naturally (no forced air applied). If the fluid is **not** pulled from the dispensing bottle, press and hold the 'Clean' button (1) on the air compressor cleaner until all 89 ml [3 fl oz] of automatic transmission fluid have been dispensed.

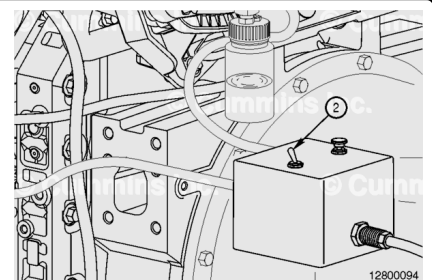
Shut the engine OFF.



Pour 89 ml [3 fl oz] of automatic transmission fluid into the dispensing bottle, and install back onto the dispensing hose assembly.

Toggle the 'Unloader' switch (2) ON to apply air pressure to the unloader signal port.

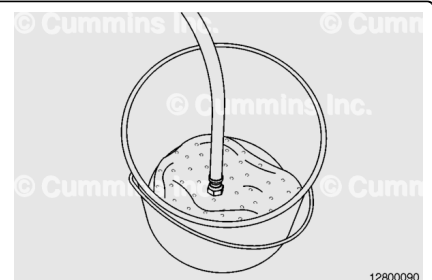
Do **not** relieve pressure at the unloader valve at this time.



Start the engine.

Immediately depress and hold the 'Clean' button on the air compressor cleaner until all 89 ml [3 fl oz] of automatic transmission fluid have been dispensed. Continue to hold the 'Clean' button down for an additional 10 to 20 seconds.

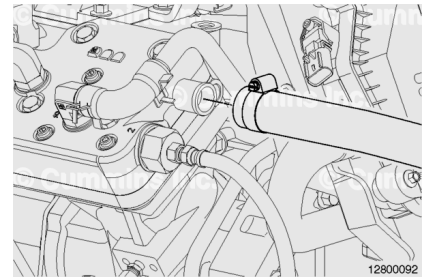
Note : A heavy mist of automatic transmission fluid should be coming from the discharge hose.



Shut the engine OFF.

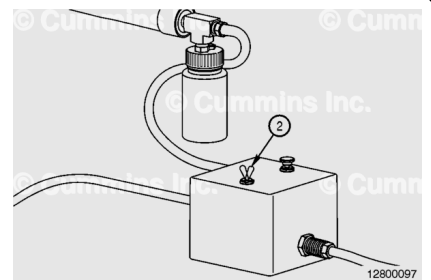
Remove the discharge hose assembly from the air compressor cylinder head intake port.

Start the engine.

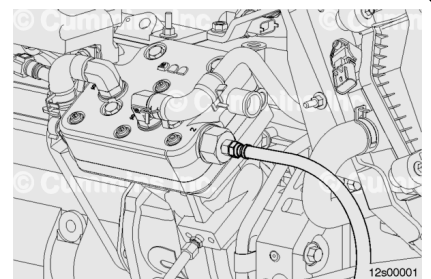


Continuously cycle the 'Unloader' switch on the air compressor cleaner for up to 25 cycles.

Note : When the air compressor cylinder head has been adequately cleaned, there should be an audible difference between loaded and unloaded states of the air compressor. Installation of a 1.5 meter [5 ft] length of rubber hose over the intake port may help to audibly identify when the unloader is functioning properly.

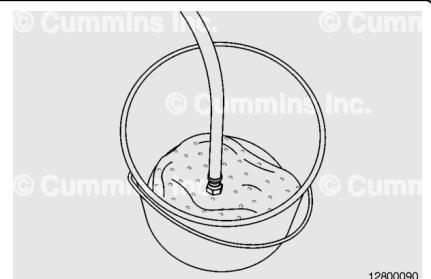


Verify constant suction at the air compressor cylinder head intake port when the air compressor is loaded (unloader switch OFF), and fluctuating air flow when unloaded (unloader switch ON). If constant suction is **not** present in a loaded state, repeat the cleaning process one additional time.



⚠ CAUTION ⚠

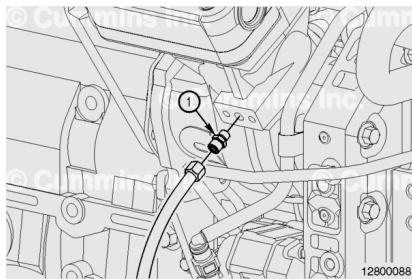
Operating the air compressor with no filter in place at the intake port may allow dust or debris to enter the air compressor. A rubber hose should be placed over the intake port and routed to an area free of air-born dust and debris.



With the discharge hose still installed, relieve the pressure at the unloader port (switch OFF), and operate the engine at high idle for 3 - 5 minutes to remove any excess automatic transmission fluid.

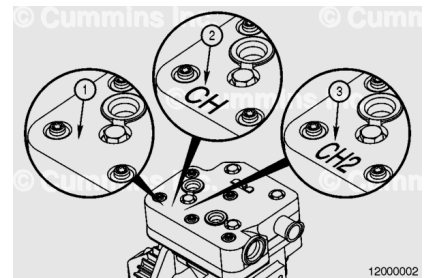
Allow engine to return to low idle. Shut the engine OFF.

Remove the unloader control valve hose and M10 connector (1).



Use an engraving pen and mark the letters CH below the head bolt as shown.

1. No marking: Perform on-engine repair.
2. CH marking: Perform off-engine repair.
3. CH2 marking: Replace the air compressor cylinder head.
Refer to Procedure 012-007 in Section 12.
(</qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html>)



Use the following procedure for a Signature™, ISX, and QSX15 engines to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.

(</qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html>)

Use the following procedure for a ISM, ISMe, and QSM11 engines to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.

(</qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html>)

Use the following procedure for a ISX15 CM2250 engine to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.

(</qs3/pubsys2/xml/en/procedures/132/132-012-014-tr-mixar5.html>)

Use the following procedure for a ISX12/11.9 CM2250 engine to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.

(</qs3/pubsys2/xml/en/procedures/133/133-012-014-tr-mixab.html>)

If the air compressor does **not** function properly, perform the Off-Engine Repair in this procedure.

Off-Engine Repair

⚠ WARNING ⚠

Use skin and eye protection when handling caustic solutions to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

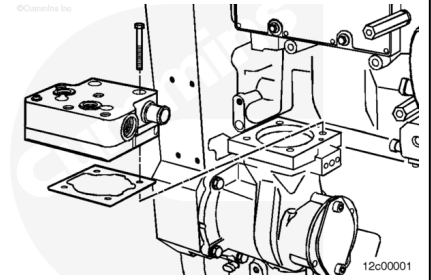
Do not use a sharp object to remove carbon. The sealing surfaces can be damaged.

⚠ CAUTION ⚠

Avoid getting debris and solvents into the clearance between the piston and bore.

⚠ CAUTION ⚠

Avoid the use of abrasive products similar to "Scotch Brite™" because any abrasive grit left after cleaning will shorten the life of the air compressor.



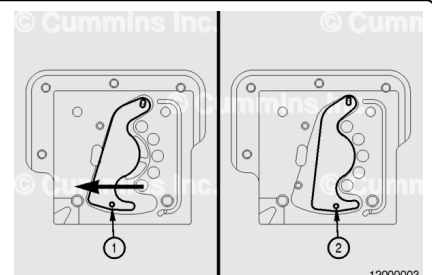
Remove the compressor head and valve assembly. Refer to Procedure 012-007 in Section 12.
(/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)

Remove carbon from the compressor head and valve assembly components with a non-abrasive solvent and a non-metallic brush.

Note : Using an abrasive solvent like carburetor cleaner can lead to further lubrication loss in the air compressor head.

Blow out passages with compressed air.

Verify the unloader slider plate is fully closed (2) when at rest.



Remove the unloader slider plate and verify proper unloader piston travel. Use a stiff wood handle, or equivalent, non-abrasive device to move the unloader piston by pushing back on the unloader slider plate guide pin in the direction of arrow as indicated (1). Be careful **not** to damage the aluminum surface of the head or drop debris into the open slot in which the guide pin travels.

Note : When released, the unloader piston should return to the loaded position (2) with a quick, smooth motion and audible snap.

If the unloader piston does **not** return completely to the loaded position with an audible snap; fill the open slot between the unloader slider plate guide pin and the closed position of the unloader position with automatic transmission fluid.

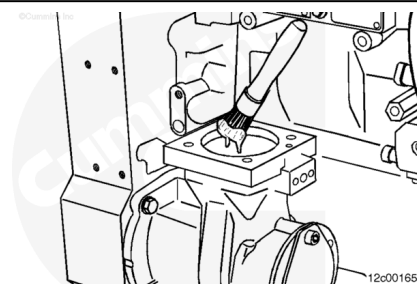
Actuate the unloader back and forth for three periods of 10 seconds. If the unloader is not free after 10 seconds, fill the open slot again with automatic transmission fluid each time until the piston is free and returns to the loaded position (2) with a quick, smooth motion and audible snap.

If the unloader piston does **not** return to the loaded position with a quick smooth motion and an audible snap when released after performing the steps above, replace the cylinder head.

Clean the compressor cylinder.

Rotate the crankshaft until the piston is at the top of the cylinder bore.

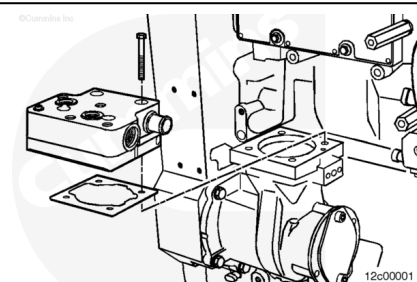
Remove any accumulated carbon and varnish by careful scraping and light application of solvents.



Using an engraving pen mark the number 2 next to the CH marking already engraved on the cylinder head.

Install the compressor head and valve assembly. Refer to Procedure 012-007 in Section 12.

(/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)



Finishing Steps

- Use the following procedure for a Signature™, ISX, and QSX15 engines to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
- Use the following procedure for a ISM, ISMe, and QSM11 engines to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html)
- Use the following procedure for a ISX15 CM2250 engine to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/132/132-012-014-tr-mixar5.html)
- Use the following procedure for a ISX12/11.9 CM2250 engines to perform the Air Compressor Diagnostic Test to verify the air compressor is functioning properly. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/133/133-012-014-tr-mixab.html)
- If the air compressor does not function properly, replace the air compressor cylinder head. Refer to Procedure 012-007 in Section 12.
(/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)

